Experiment 4: Crypt-Arithmetic Problem

Aim

To solve a Crypt-Arithmetic problem using Python.

Objective

To find the digit values for letters in a Crypt-Arithmetic puzzle such that the arithmetic equation is satisfied.

Algorithm

1. Import the `itertools` module.
2. Define the words **SEND**, **MORE**, and **MONEY**.
3. Generate all possible digit assignments for the letters.
4. Ensure that no two letters have the same digit.
5. Check if the equation **SEND + MORE = MONEY** is satisfied.
6. If the equation is true, display the solution.
7. Stop the program.

Flowchart

```
          Start
           |
           ▼
Import itertools Module
           |
           ▼
Assign Digits to Letters
           |
           ▼
Are Digits Unique?
   |       |
   No      Yes
   |       |
   ▼       ▼
 Try Next   Check Equation
            |
            ▼
     Is SEND + MORE = MONEY?
        |          |
        No         Yes
        |          |
        ▼          ▼
     Try Next     Display Solution
                   |
```

Stop

Python Program

```
from itertools import permutations

letters = ('S','E','N','D','M','O','R','Y')

for p in permutations(range(10), 8):
    S,E,N,D,M,O,R,Y = p

    if S == 0 or M == 0:
        continue

    send = 1000*S + 100*E + 10*N + D
    more = 1000*M + 100*O + 10*R + E
    money = 10000*M + 1000*O + 100*N + 10*E + Y

    if send + more == money:
        print("SEND =", send)
        print("MORE =", more)
        print("MONEY =", money)
        break
```

Sample Output

Result

Thus, the Crypt-Arithmetic problem was solved successfully using Python.

Conclusion

The program assigns unique digits to each letter and checks all possible combinations until it finds a valid solution satisfying the equation **SEND + MORE = MONEY**. It demonstrates the use of **permutations** and **constraint satisfaction** in Artificial Intelligence.